

SECTION B

Answer **all** questions.

6. Read through the following article carefully.

How do you produce really low temperatures and can you then do some cool stuff with it?

Paragraph

Pressure / 10^5 Pa

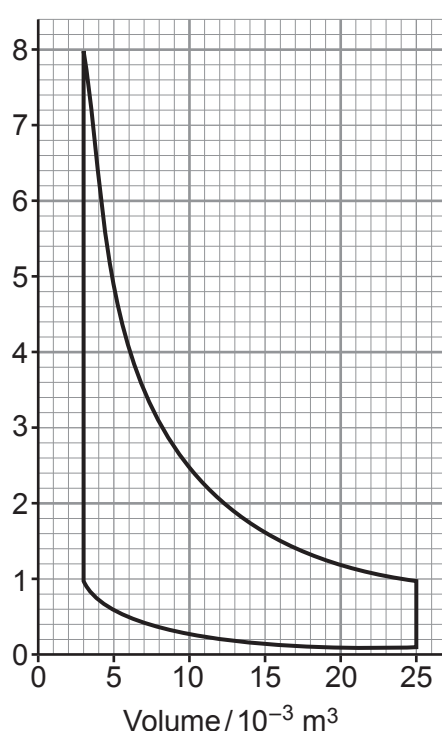


Figure 1

The first thing you have to do is to produce low temperatures and this is usually achieved by taking a gas through a closed cycle such as that shown in Figure 1.

1

1. Air, initially under high pressure, is expanded at constant temperature.
2. Heat is allowed to escape at constant volume.
3. The gas is compressed at a constant low temperature.
4. The gas is heated at constant volume to a high pressure.

2

The end result is that the gas does a large amount of work in each cycle. This means that the gas must have heat flowing into it leading to the cooling of whatever it is that requires cooling.

3

A different and simpler method of cooling gases is called the Joule-Thompson method, which does not involve a gas going through a cycle. A gas is compressed to a pressure of two thousand times atmospheric pressure. The gas is then allowed to cool back to room temperature. After that, it is allowed to expand quickly through a nozzle to atmospheric pressure leading to a tremendous amount of further cooling. In the expansion, the gas does a very large amount of work but the non-ideal gas molecules also have an enormous increase in potential energy because of the attractive forces between molecules. Both these factors cool the gas by a huge factor.

4

Okay, so now we have the basic physics for producing low temperatures. Apply this low temperature to air and we can make a product that can give hours of fun – liquid nitrogen (LN_2).

5

Most of the fun that can be obtained from LN_2 is based on the fact that it boils at a temperature of -195.8°C and, in doing so, expands by a factor of approximately 1 000. For example, a cool rotating rocket can be built using a simple plastic water bottle.

6



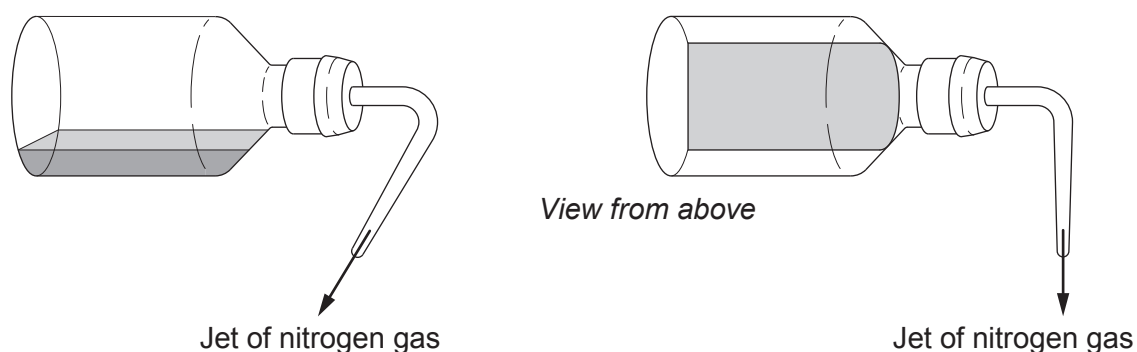


Figure 2

Paragraph

The boiling nitrogen makes the nozzle of the plastic water bottle into a jet engine. When you then place the plastic water bottle on a nicely polished wooden floor it will rotate far more quickly than you would ever anticipate, providing a great demonstration for even the most ardent physics-hating pupil. What can be even more spectacular is when the bottle accidentally hits something solid (like the leg of a lab desk). The wall of the plastic water bottle will be at around -200°C and fragile. This, combined with the high pressure inside the bottle, leads to rather a loud and impressive bang.

7

Although LN_2 can be fun, it can also be dangerous. For instance, when transporting a large, open container of LN_2 (see Figure 3), you should not remain in the confined space of a lift with the container in case the lift becomes stuck. Most of the danger from LN_2 arises from its 200°C temperature difference from its surroundings. Nonetheless, it is always amusing when TV chefs wear safety goggles and gloves when making ice cream using LN_2 but never bother with any safety equipment when using litres of boiling fat above a gas fire. I promise you that the boiling fat is far more dangerous even though the temperature difference, compared with human body temperature, is around 200°C for both LN_2 and boiling fat.

8

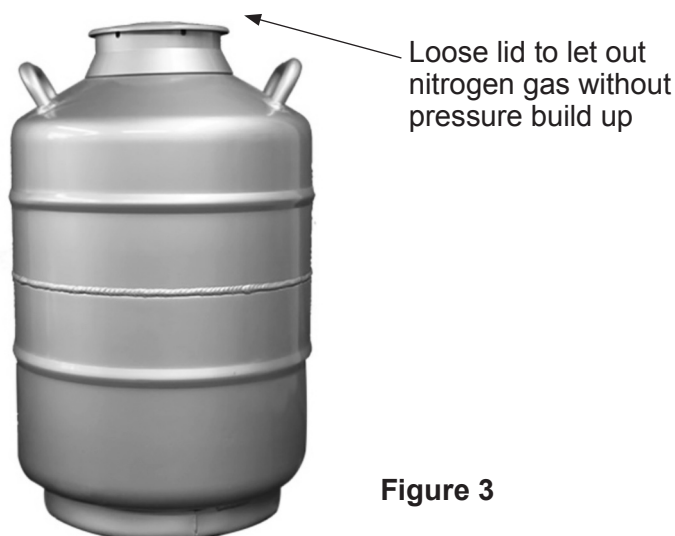


Figure 3

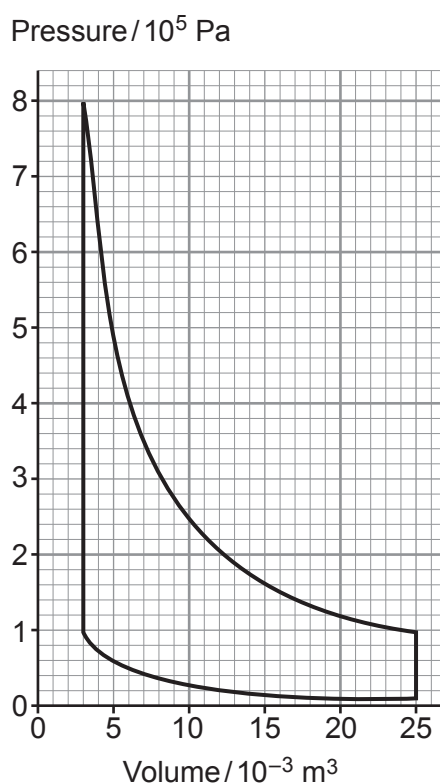
In conclusion, low temperatures and LN_2 can be dangerous for some obvious reasons but they can also lead to some amusement and even some Michelin-starred ice cream.

9



Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

- (a) **Label the diagram** with the processes 1–4 along with an arrow to show the direction of the cycle (see paragraph 2). [2]



- (b) The author states that work is done by the gas in the cycle. By considering the four stages of the cycle, explain whether or not this statement is correct (see paragraph 3 and Figure 1). [2]

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(c) Estimate the work done during the cycle.

[3]

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(d) When a gas is compressed in a metal cylinder it will become hot. Explain why the gas will then cool to room temperature (see paragraph 4).

[2]

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(e) Explain whether or not the author is correct to state that the gas expanding quickly through the nozzle will cool greatly (see paragraph 4).

[4]

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(f) Explain very briefly why nitrogen gas escapes through the nozzle of the plastic water bottle (see Figure 2 and paragraph 7).

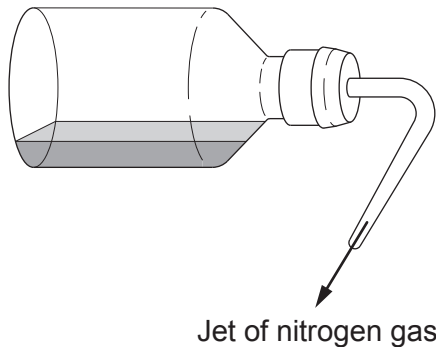
[1]

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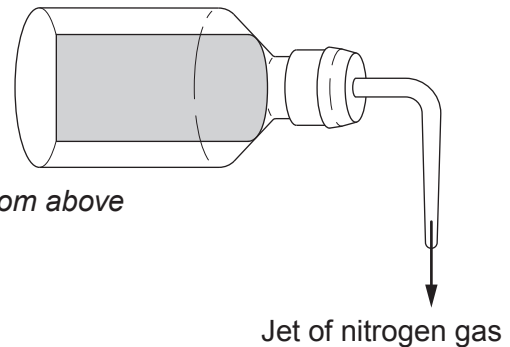
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- (g) Starting from Newton's 3rd Law, explain why the plastic water bottle shown will rotate anti-clockwise. [2]



View from above



- (h) An exploding bottle produces a "loud bang" (see paragraph 7). Explain briefly why this "loud bang" can be heard more than 100 m from the exploding bottle. [2]

- (i) Justify the author's statement that it would be dangerous to share a confined space (like a lift) with a large container of liquid nitrogen (see paragraph 8 and Figure 3). [2]

END OF PAPER

